

Shubh Somani

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Summary — Aspiring Machine Learning Engineer , Passionate About AI and Innovation , Data Structures & Algorithms Learner , Computer Vision Implementation in real life cases .

Skills

Languages Python, Batch, Basics C , Java , IOT
OS Windows , Debian(Kali), RedHat(Fedora),
TrueNAS(HomeLaging), Azure Cloud

Libraries Docker ,Pytorch, Tensorflow, Scikit-Learn,
LangChain, Ultralytics, MediaPipe, OpenCV,
Numpy, Pandas, Matplotlib

Experience

IEEE CS (Computer Society)

Oct 2024 – Apr 2026

Junior Working Team (Technical)

Team Project : Virtual Mentor

- AI-powered learning assistant that leverages the capabilities of LLM's to provide students with an intelligent academic companion.
- Integrating multi-modal RAG to process and understand course materials across all eight semesters of study, offering precise, contextual, and interactive responses.

Core Working Team (Technical)

Team Project : Hiring Assistant

- AI-powered hiring platform that automates resume screening and candidate ranking using NLP, embeddings, and cosine similarity, reducing manual recruiter effort and shortlisting time.
- A scalable web system using FastAPI and llama-based models to extract skills, match resumes with job descriptions, and generate AI-driven interview questions through a modular REST architecture.

Education

GD Goenka Public School

CBSE 12th (PCM)

CBSE 10th

Certifications

- Stanford - Unsupervised Learning, Recommenders, Reinforcement Learning
- Stanford - Supervised Machine Learning: Regression and Classification
- IBM - Python for Data Science, AI & Development (IBM)

Projects

Plant Disease Classification with Noise Augmentation

Jan 2026

- Multi-model deep learning system for automated plant disease detection using ResNet50, DenseNet201, EfficientNet, and YOLOv8 architectures on 27-class PlantDoc dataset.
- Implemented Gaussian noise augmentation techniques and comprehensive model comparison framework with performance metrics analysis to improve classification robustness.

Manipal ChatBot (CampusPal)

Apr 2025

- A special large language model created using the RAG system and fine-tuning to relay Manipal University-specific responses to the chatbot.
- Upload your own PDF content to ask questions about your substance.
- All of our university's notes are located in one place.

Vaidya-Health-Care-with-Yoga-Assistant

Jan 2025

- Comprehensive healthcare management system that includes various applications for yoga pose tracking, disease prediction, mental health analysis, and more.
- The system leverages machine learning models, computer vision, and natural language processing to provide real-time feedback and insights.

Self-Hosted Home Cloud & Homelab Infrastructure (TrueNAS SCALE)

Dec 2025

- Built and maintained a self-hosted home cloud on TrueNAS SCALE using mirrored storage (RAID1) on enterprise hardware, deploying multiple Docker containers and VMs including media servers, automation tools, DNS filtering, and a game server.
- Implemented secure remote and public access using Tailscale, Cloudflare Tunnels, and NGINX Reverse Proxy, gaining hands-on experience with networking, container orchestration, storage management, virtualization, and self-hosted DevOps workflows.